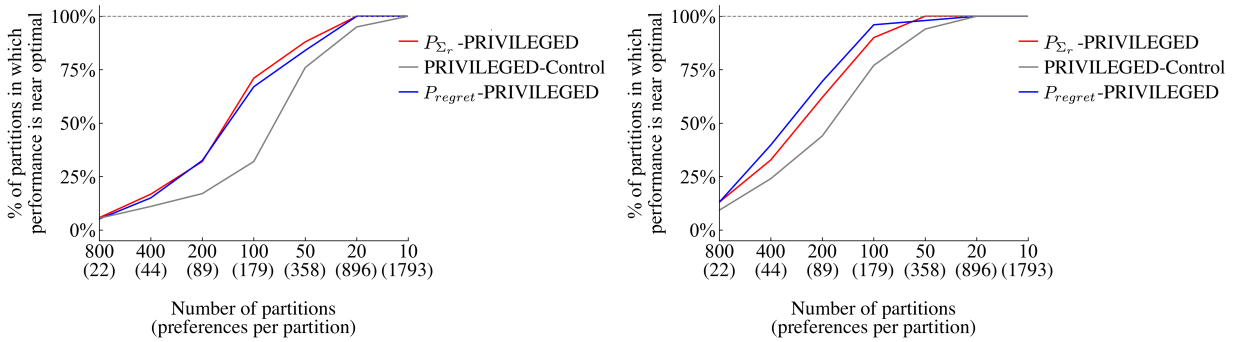

Figure 6: Learning a reward function with the partial return preference model (Left) and regret preference model (Right) from the preferences collected in the PRIVILEGED experiment. Each preference dataset is randomly partitioned into equal-sized training sets 10 times, using 10 different random seeds to control partitioning. We learn a reward function using the given preference model for each training set, and plot the percent of partitions—or equivalently training sets—in which the learned reward function induces near-optimal behavior. Note this percentage is across all partitions across all 10 seeds. To visually test for Hypothesis 2, observe the gap between the colored (i.e., red and blue) lines and the gray line.

Results are shown in Figure 6, with reward learning details outlined in Appendix J for all experiments. Appendix J additionally presents results comparing performance of the resultant policy to that of a uniformly random policy (V_r^U).

Figure 6 supports Hypothesis 2. Learning with the partial return preference model from the P_{Σ_r}-PRIVILEGED dataset results in reward functions that induce near-optimal behavior more often than when learning from the PRIVILEGED-Control dataset for all partition sizes, except for the largest partition size where performance is matched. A comparable pattern is observed when learning from the P_{regret}-PRIVILEGED dataset with the regret preference model.

5.2 TRAINED Experiment

In real-world practice, RLHF practitioners do not have access to the ground-truth reward function during preference elicitation. Therefore, in this experiment and the next, we focus on interventions to improve preference model alignment that are feasible in practice. In the TRAINED experiment, we evaluate training humans to follow either the partial return or regret preference models during preference elicitation. All experimental details are the same as those of the PRIVILEGED experiment unless otherwise noted.

This experiment consists of three conditions. In the P_{Σ_r}-TRAINED and P_{regret}-TRAINED intervention conditions, subjects are taught to follow the corresponding preference model. In the TRAINED-Control condition, subjects are not taught about any specific preference model and are trained identically as all subjects in the PRIVILEGED experiment. The conditions differ only by how subjects are trained. Subjects are randomly assigned to a condition until each condition has data from 10 subjects. Unlike in the PRIVILEGED experiment, if a subject's preference data is removed because of poor task comprehension or inattentiveness (see Appendix C.4), their slot is reopened for random assignment, including their exact set of segment pairs to be labeled with preferences. With this replacement strategy, we gather preferences between the same set of segment pairs for each condition, which produces paired data. We discuss this design choice in Appendix H. A video walk-through of the interface used for the P_{Σ_r}-TRAINED condition is available here, for the P_{regret}-TRAINED condition here, and for the TRAINED-Control condition here.

Intervention details: subject training During training, subjects in the intervention conditions learn about the domain's dynamics and reward function, as well as the segment statistic specific to their condition (partial return or regret). They are shown the segment statistic while interacting with the delivery domain and taught how to compute it. We then provide subjects with a detailed example of how to use the taught segment statistic to generate preferences, have them practice with feedback, and finally ask them to label preferences over segment pairs from the delivery task. Subjects label preferences for 50 segment pairs with the elicitation question changed based on the condition. Further details on the training protocol are provided in Appendix E.

Hypothesis 1: Training a human to follow a specific preference model will influence the human to give preferences according to that model.

We follow the procedure outlined in Section 5.1 to evaluate evaluate Hypothesis 1 with results shown in Figure 7. Hypothesis 1 is supported; the intervention condition datasets are more likely under the respective target preference model than the control condition. Because samples are paired, we perform a Wilcoxon paired signed-rank test comparing the likelihood of samples in the control dataset to those in the dataset from training humans to follow a preference

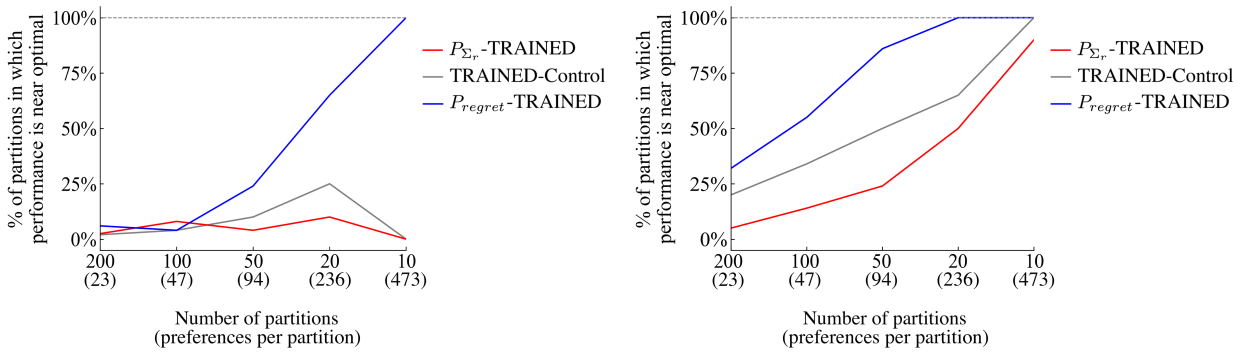

Figure 8: Learning a reward function with the partial return preference model (Left) and regret preference model (Right) from the preferences collected in the TRAINED experiment. See Figure 6 for more details on how this figure was generated.

model, finding a statistically significant difference at $p < 0.01$ that further supports Hypothesis 1. See Appendix H for details. Additionally, we find that the accuracy over both the P_{Σ_r}-TRAINED and P_{regret}-TRAINED datasets is notably higher than the accuracy over the TRAINED-Control dataset given the noiseless version of the respective target preference model. These results, detailed in Appendix I, additionally support Hypothesis 1 with statistical significance at $p < 0.01$.

Hypothesis 2: Training humans to follow a specific preference model leads to learning more aligned reward functions with that preference model.

Figure 8 illustrates partial support for Hypothesis 2. Learning from the P_{regret}-TRAINED dataset results in reward functions that induce near-optimal performance more often than learning from the TRAINED-Control dataset when using the regret preference model for learning. But, learning from the P_{Σ_r}-TRAINED dataset when using the partial return preference model leads to comparatively poor performance. See Appendix J.6 for a discussion on this result.

For larger training sets, learning a reward function with the partial return preference model using the P_{regret}-TRAINED dataset instead of the other condition's datasets results in reward functions that induce near-optimal behavior more often; influencing preferences towards P_{regret} appears to be generally beneficial for reward learning in this task.

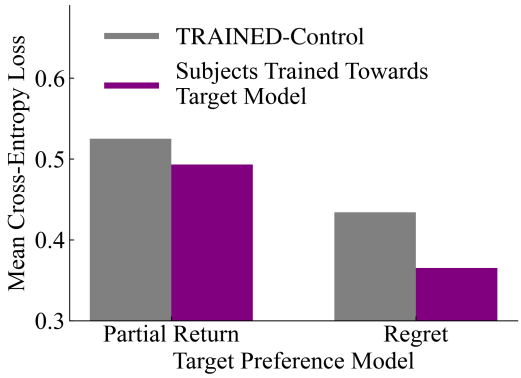

Figure 7: For the TRAINED experiment, mean cross-entropy loss over each condition's preference dataset with respect to the target preference model. Lower is better. Performing a Wilcoxon paired signed-rank test results in $p < 0.01$ for both conditions.

Additionally, in Appendix E, we provide results indicating that subjects generally agreed with how the taught preference model generated preferences, but find no statistically significant correlation between subjects' ability to compute the relevant segment statistics and their adherence to the target preference model during preference elicitation.

5.3 QUESTION experiment

For some domains, it may be difficult or time consuming to teach humans about a specific preference model. Concepts such as the partial return of a trajectory segment may be difficult for humans to comprehend in other environments, for example when eliciting preferences over large language model outputs. Instead, we change only the wording of the question asked during preference elicitation, investigating how to align human preferences with specific preference model without relying on their explicit understanding of the preference model.

This experiment consists of three conditions; the P_{Σ_r}-QUESTION and P_{regret}-QUESTION conditions, where we change the preference elicitation instruction in favor of the partial return preference model and regret preference model, respectively, and the QUESTION-Control condition where we use a preference elicitation question that does not seek to influence human preferences towards any specific preference model. Subjects' training matches that of the previous experiment's control condition, and each condition differs only in the wording of the question we ask when eliciting preferences from subjects. We collect data from 9 subjects per condition who are assigned to conditions via random assignment. We replace the data of subjects that are removed due to poor task comprehension or inattentiveness. A

video walk-through of the interface used for the P_{Σ_r}-QUESTION condition is available here, for the P_{regret}-QUESTION condition here, and for the QUESTION-Control condition here. Appendix F contains further details.

Intervention details: Preference elicitation questions The subjects in each condition see the corresponding question below:

- QUESTION-Control - "Which path do you prefer?", chosen to reduce influence
- P_{Σ_r}-QUESTION - "Which path has better immediate outcomes?", chosen to focus subjects only on the reward within a segment
- P_{regret}-QUESTION - "Which path reflects better decision-making?", chosen to reflect regret's measurement of a segment's deviation from optimality

Hypothesis 1: Changing the preference elicitation question in favor of a specific preference model will influence the human to give preferences according to that model.

Results are shown in Figure 9, supporting Hypothesis 1 with a relatively small effect size. The loss for the P_{Σ_r}-QUESTION dataset under the partial return preference model is lower than the loss for the QUESTION-Control dataset, indicating a small shift in subject's preferences towards the partial return preference model. A Wilcoxon paired signed-rank test between the likelihoods of these two datasets indicates a statistically significant difference at $p < 0.05$. The P_{regret}-QUESTION dataset's loss under the regret preference model is also lower than, but close to, that of the QUESTION-Control's dataset with no statistically significant difference. We observe a similar pattern when looking at the accuracy of the noiseless target preference model over the P_{Σ_r}-QUESTION and P_{regret}-QUESTION datasets with similar significance. See Appendix H and I for more details.

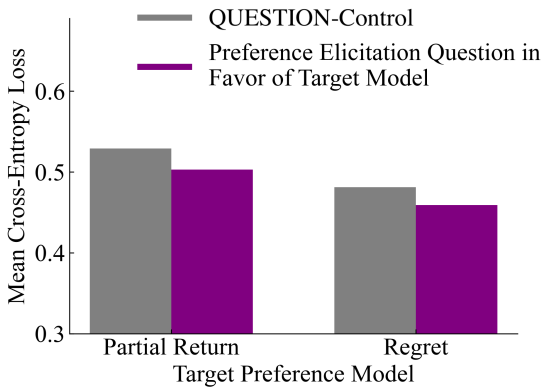

Figure 9: For the QUESTION experiment, mean cross-entropy loss over each condition's preference dataset with respect to the target preference model. Lower is better. Performing a Wilcoxon paired signed-rank test results in significance for only the P_{Σ_r}-QUESTION condition ($p < 0.05$).

Hypothesis 2: Changing the preference elicitation question in favor of a specific preference model leads to learning more aligned reward functions with that preference model.

Figure 10 provides evidence in support of Hypothesis 2. Modifying the preference elicitation question to steer human preferences towards a specific preference model results in a dataset that induces near-optimal behavior more often—when learned from using the target preference model—compared to the control condition dataset. Broadly speaking, this suggests that the question we ask subjects when labeling preferences can effect the performance of the resulting learned reward function.

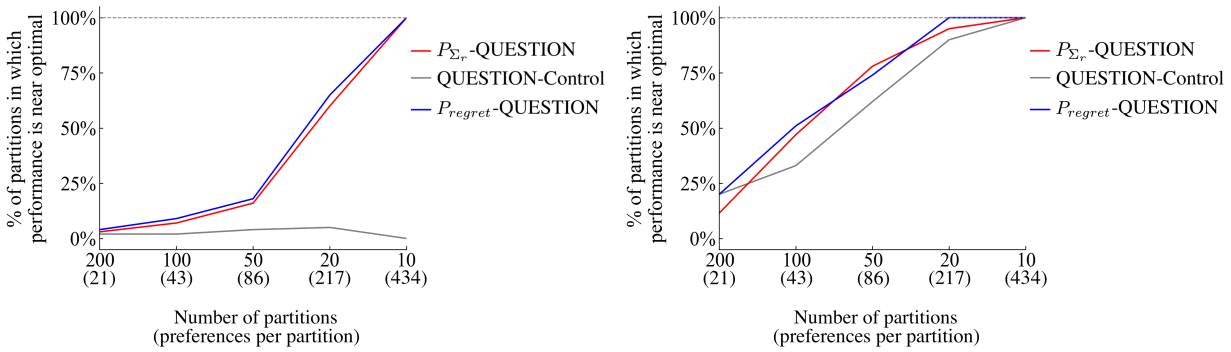

Figure 10: Learning a reward function with the partial return preference model (Left) and regret preference model (Right) from the preferences collected in the QUESTION experiment. See Figure 6 for more details on how this figure was generated.